

Student Onboarding Automation

- **Executive Summary**

Currently, the university manages the student onboarding process through a combination of manual procedures, spreadsheets, email communications, and disparate systems. While these systems have previously supported onboarding activities, they are becoming progressively ineffective and difficult to manage as student enrollment volumes increase.

The current procedure requires extensive coordination among several departments, including admission, finance, and academic services. The reliance on manual intervention frequently results in processing delays, communication inconsistencies, duplication efforts, and lack of real-time visibility into the status of student onboarding activities across departments.

To overcome these issues, the university recommends implementing a Student Onboarding Automation System. The proposed solution will provide a digital platform that automates and simplifies onboarding activities from admission acceptance through student activation. It aims to increase operational efficiency, improve the student experience and reduce administrative workload across all participating departments.

- **Problem Statement:**

The current student onboarding process relies heavily on manual effort, disconnected communication methods, and separate administrative systems that do not effectively share information. As a result, university employees struggle to coordinate operations across departments while newly admitted students frequently face delays in completing onboarding requirements.

Key issues identified within the existing process include the manual submission, review, validation of student documentation, heavy reliance on email-based communication, repetitive data entry and absence of centralized onboarding tracking system. These challenges result in duplication efforts, increasing process

time, communication inefficiencies, increasing the risk of errors and data inconsistencies.

- **Project Overview:**

The Student Onboarding Automation Project aims to develop a centralized digital platform that support the onboarding journey for newly admitted students. The proposed solution will help students with self-service portal as it will allow them complete onboarding activities, including, account creation, document submission, status tracking, payment, and communication with related university departments. It will also provide automated notifications and real-time updates throughout the onboarding process.

From an administrative perspective, the solution will enable the university departments to manage onboarding activities. Admission, finance, and academic services will be able to review submissions, process approvals, track progress and create student record through a unified platform.

- **Business Objectives**

The primary objective is to establish efficient onboarding process by implementing digital platform. The project aims to achieve the following business objectives:

- 1- Improve operational efficiency.
- 2- Reduce manual data entry and human errors.
- 3- Reduce onboarding cycle time.
- 4- Enhance student experience.
- 5- Enhance communication and stakeholder engagement.
- 6- Increase process visibility and transparency.
- 7- Improve data accuracy and consistency.
- 8- Strengthen cross-departmental collaboration.
- 9- Enhance reporting and decision making capabilities.

- **Scope**

In scope:

- **Digital student onboarding platform:**
 - Provide a centralized portal that enables newly admitted students to complete onboarding activities, submit required information, upload documents, and monitor onboarding progress.
- **Automated onboarding processes:**
 - Automate onboarding workflows, approvals, validations, and related onboarding activities across relevant university departments.
- **Electronic document management:**
 - Enable the digital submission, review and secure storage of student onboarding documents.
- **Automated communication and notifications:**
 - Provide automated notifications and status updates to ease communication between students and university staff.
- **Integration with existing university systems:**
 - Integrate the onboarding platform with existing ERP, student information system, finance, and other relevant university systems to facilitate seamless data exchange and reduce manual data entry.
- **User access management and monitoring dashboards:**
 - Allow students, staff, and managers to access the system based on their roles and provide dashboards to view relevant information and track progress.
- **Management reporting and business insights:**
 - Generate operational reports, onboarding statics and management dashboards to support monitoring and decision-making.

Out of scope:

- Core ERP replacement:
 - The project will integrate with existing ERP system but will not replace or modify core ERP functionalities.
- Learning management system:
 - Courses delivery, e-learning, assessments and examinations not included within the scope of the project.
- Alumni management system:
 - Alumni records and post-graduate services outside of the scope.
- Course Registration and scheduling:
 - Course enrollment, timetable management and academic scheduling processes are not included.
- Graduation and certification processes:
 - Graduation requirements and certificate management are not included.

• Stakeholders

Stakeholder	Description	Role
Students	Newly admitted students who use the onboarding system (end users).	Submit required information and documents, track onboarding status.
Admission Department	Admission and registration team.	Review admission records and validate submitted documents for admitted students.
University Management	University leaders.	Provide guidance, approve key decisions, and ensure the project supports university objectives.
Finance Department	Department responsible for handling student financial	Check tuition payments, confirm payment and update payment status during onboarding process.

	transactions and payment validation.	
IT Department	Team responsible for supporting and maintaining university systems.	Manage system implementation, integration, security, maintenance and technical support.
Software Development Team	Team responsible for delivering the project and coordinating implementation activities.	Manage project requirements, design, development, testing, training, deployment, and overall project delivery.

- **As-Is Process**

The current student onboarding process is manual and involves multiple departments operating through separate systems, emails, and spreadsheets.

- **The current onboarding process flow:**

- 1- Student submit required documents by email or visiting the university.
- 2- Admission department reviews and validates the submitted documents.
- 3- If there are any issues the admission department contacts the student via email and request the missing documents.
- 4- The student resubmits the missing documents.
- 5- The finance department manually verifies fee payment.
- 6- The academic service manually creates and updates the student record in the system.
- 7- Departments track onboarding process using emails, spreadsheets, and separate systems.
- 8- The student follows up with different departments through emails to track the status updates.

9- Once the system meets the requirements, the onboarding process is finalized and the student is officially enrolled.

▪ **Current challenges:**

- Heavy reliance on manual communication and administrative activities.
- Repeated data entry across records and multiple systems.
- Time-consuming processing and approval procedures.
- Limited visibility of onboarding progress and status.
- Higher risk of human errors and data inconsistencies.
- Possibility of missing documents or incomplete submissions.
- Lack of communication and coordination among departments.
- Limited capabilities for reporting and performance monitoring.

● **To-Be Process:**

The proposed solution will provide a centralized and automated onboarding experience for students and university staff.

▪ **The future onboarding process flow:**

- 1- The student creates account and logs in the student onboarding portal.
- 2- The student completes the required information and uploads the required documents through the portal.
- 3- The system validates the submitted information and verifies that all required documents have been submitted.
- 4- If any information or documents are missing, the system notifies the student.
- 5- The admission department review and approves the submitted documents.
- 6- The finance department verifies the fee payment and update the payment status.
- 7- The academic service creates or activate the student record and completes the enrollment process.

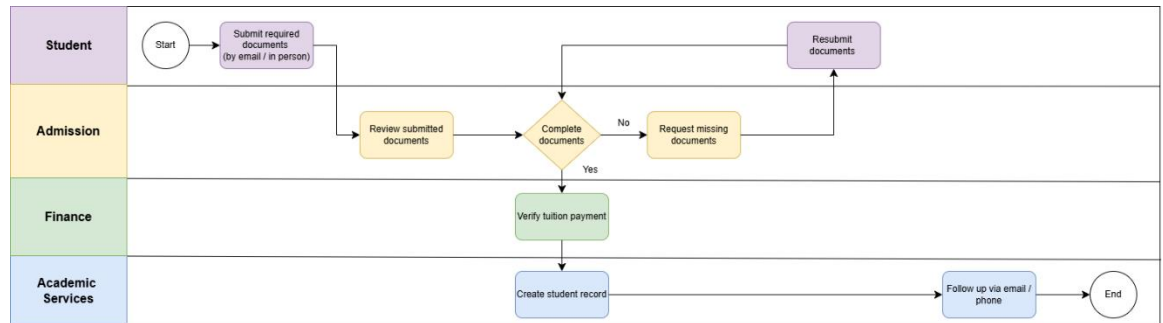
- 8- Approved student information is automatically synchronized with ERP and related university systems.
- 9- Enrollment, financial, administrative records are updated automatically across integrated platforms.
- 10- The system updates the onboarding status and provides real-time visibility to relevant stakeholders.
- 11- The student receives automated notification and status updates throughout the process.
- 12- The student can monitor the onboarding progress through the portal dashboard.
- 13- Once all requirements are completed, the onboarding process is finalized and student is officially enrolled.

■ **Expected Benefits:**

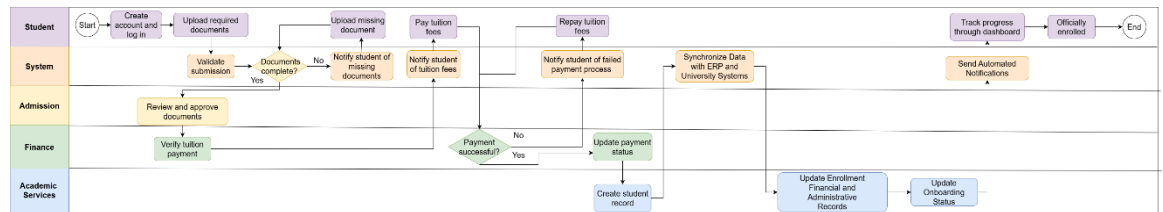
- Reduced onboarding processing time and administrative effort.
- Improved operational efficiency through workflow automation.
- Enhanced student experience through self-services capabilities and real-time updates.
- Increased visibility and transparency across the onboarding process for all stakeholders.
- Improved data accuracy and consistency across departments.
- Strengthened collaboration and information sharing between university departments.
- Enhanced reporting, monitoring, and decision-making capabilities.

- **BPMN Diagrams:**

- **As-In-BPMN:**



- **To-Be-BPMN:**



- **Functional Requirements:**

Requirement
Student should create an account using admission credentials
Student should upload required documents
System should validate uploaded documents
System should notify students of missing documents
Admissions staff should review documents and provide approval
Finance should verify payments
Academic service should activate student profile
Student should track onboarding status
System should sent automated notifications
System should generate onboarding reports
System should provide dashboards for administrators
System should support role-based access control

- **Non-Functional Requirements:**

Requirement Category	Description
Availability	The system should maintain a minimum availability of 99% during operational hours.
Performance	The system should response in 3 seconds under normal operating conditions.
Capacity	The system should concurrent users during peak admission and enrollment periods without degradation in performance.
Security	The system should provide secure user authentication to ensure authorized access to system functions and data.
Data protection	Data should be encrypted during transmission and storage in accordance with the university's information security standards.
Usability	The system should provide the a user-friendly interface.
Scalability	The system should support the future growth in student enrollment volumes without significant system redesign.

- **Assumptions:**

- Students have access to the internet and devices to use the student onboarding portal.
- Student possess valid admission credentials.
- Existing university systems (ERP, SIS, and Finance) support the required integrations.
- Student data can be migrated.
- Required technical infrastructure and system resources are available.
- Required budget and resources for the project are provided.

- **Risks:**

Risk	Solution
User resistance to change	Training and awareness sessions
Integration issues	Early integration testing
Data migration errors	Conduct migration testing
Incomplete requirements	Regular stakeholder reviews
System downtime	Backup plan
Performance issues	Performance testing and capacity before deployment.

- **Constraints:**

- Budget limitations.
- Fixed project timeline aligned with admission cycles.
- Mandatory integration with existing ERP, SIS, and Finance systems.
- Compliance with University policies and data privacy regulations.
- Adherence to existing technology infrastructure and standards.
- Limited availability of business and technical resources.

- **Success Metrics (KPIs):**

KPI	Target
Onboarding time	Reduce by 50%
Manual emails	Reduce by 80%
User satisfaction	Above 90%
Data errors	Reduced by 70%